

Ans. to the ques. no. 1

a) Every 1 is followed by odd number of 0's

$$0^* (10(00)^*)^*$$

b) Doesn't begin with 110

$$11(1)^+ \Sigma^* \cup 10\Sigma^* \cup 0\Sigma^*$$

c) Contain substring 110 but 00

$$(101 \cup 01 \cup 1)^* (110)^+ (1)^+ (101 \cup 01 \cup 1)^*$$

d) Contains 1 in every 4th position

$$(\Sigma\Sigma\Sigma 1)^*$$

e) Doesn't contain exactly 3 0's.

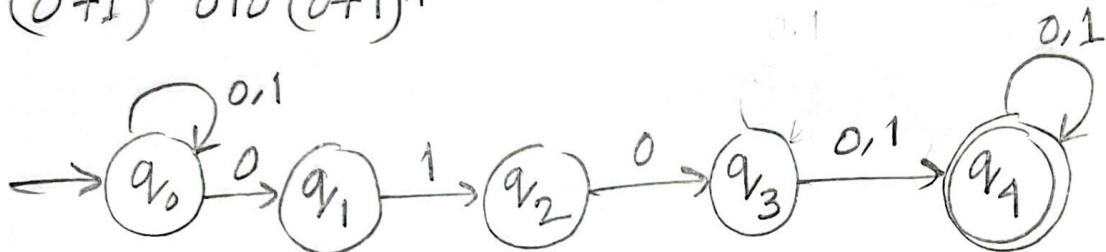
$$1^* 01^* \cup 1^* (00)^* 1^* \cup \Sigma^* (0000)^+ \Sigma^*$$

f) Has the symbol it starts with in every even position.

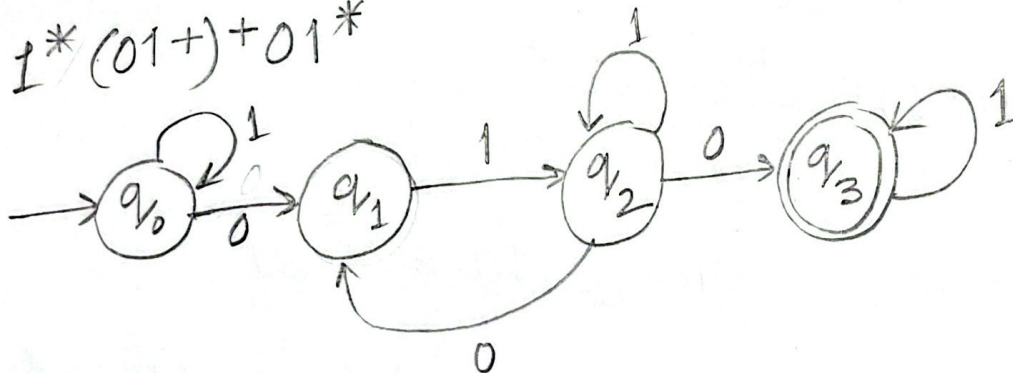
$$11(\Sigma 1)^* \Sigma \cup [00(\Sigma 0)^* \Sigma]$$

Ans. to the ques. no. 2

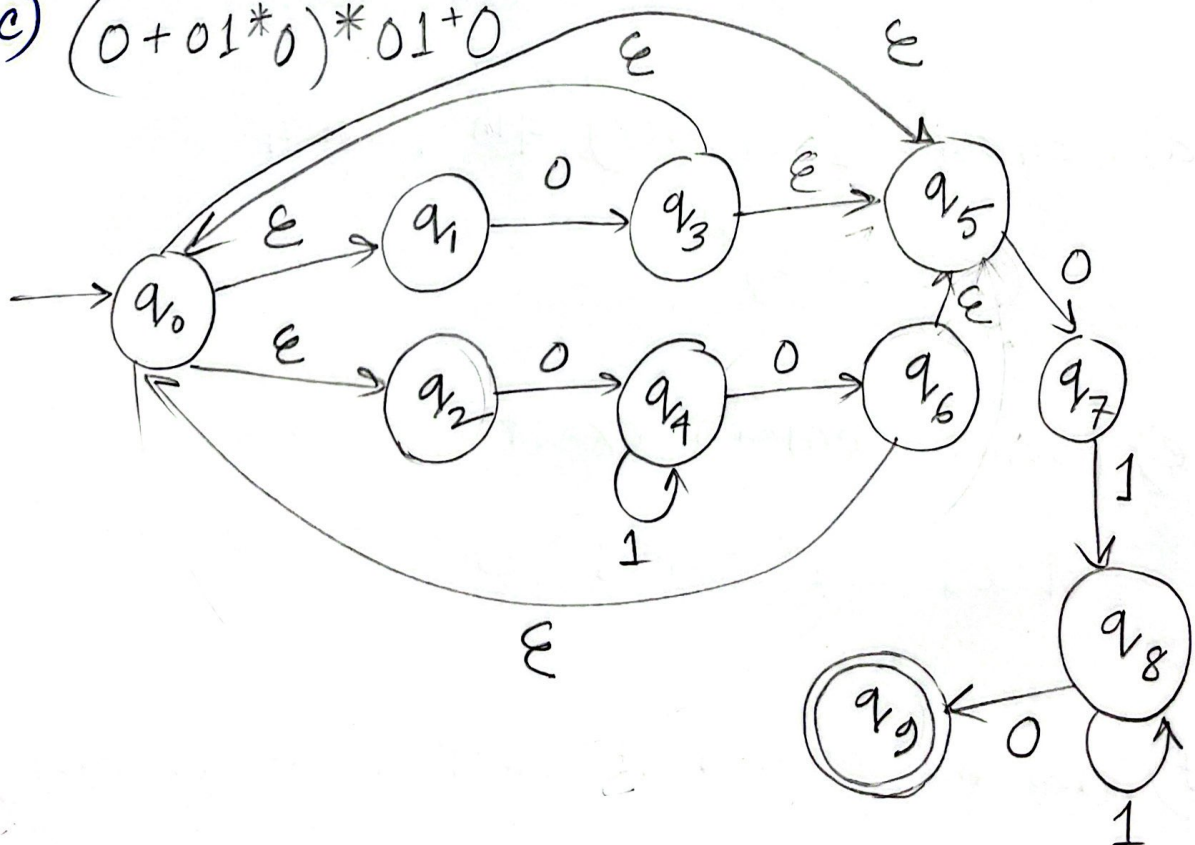
a) $(0+1)^*010(0+1)^+$



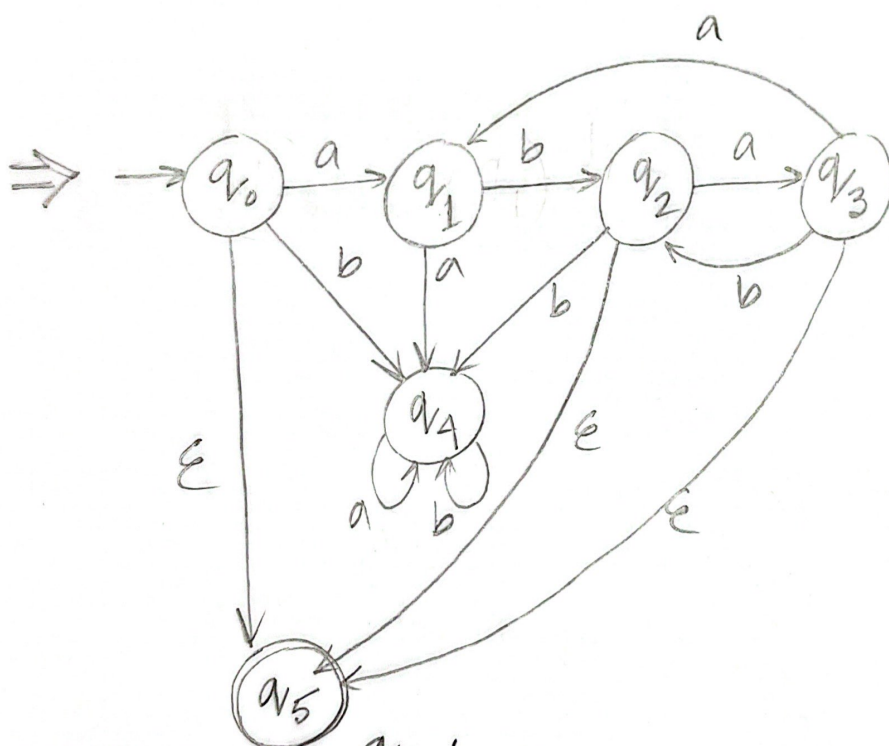
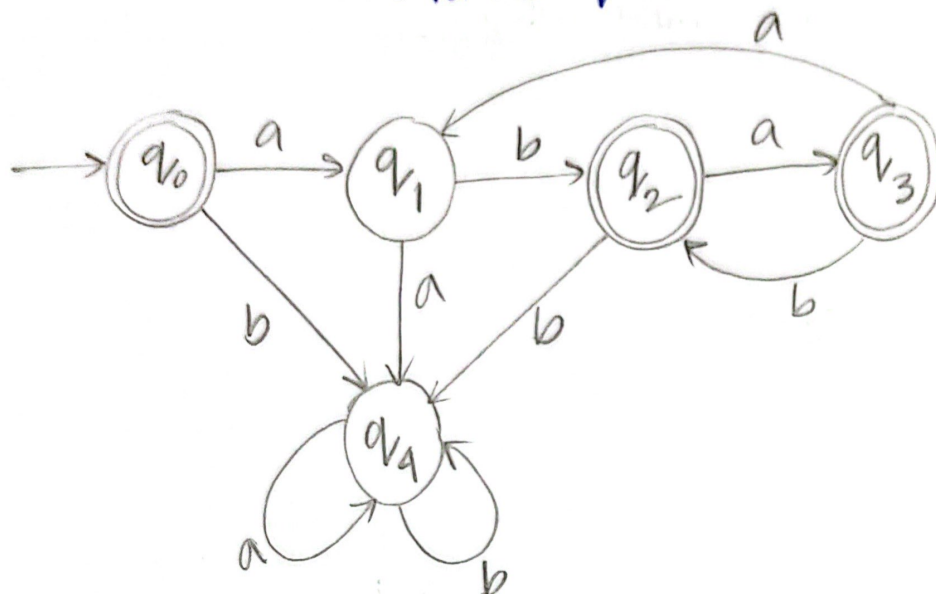
b) $1^*(01+)^+01^*$



c) $(0+01^*0)^*01^+0$



Ans. to the ques. no. 3



eliminating q_2 :
Incoming edges to q_2 & outgoing edges:

$$(q_1, q_3) \times (q_3, q_4)$$

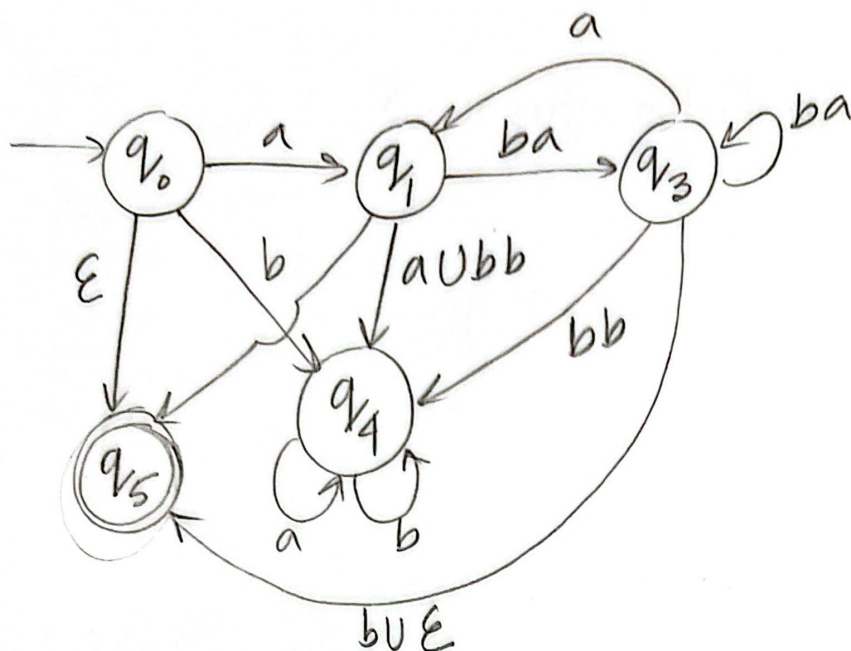
$$= (q_1, q_3), (q_1, q_4), (q_3, q_3), (q_3, q_4)$$

$$q_1 \rightarrow q_3 : ba$$

$$q_1 \rightarrow q_4 : bb$$

$$q_3 \rightarrow q_3 : ba$$

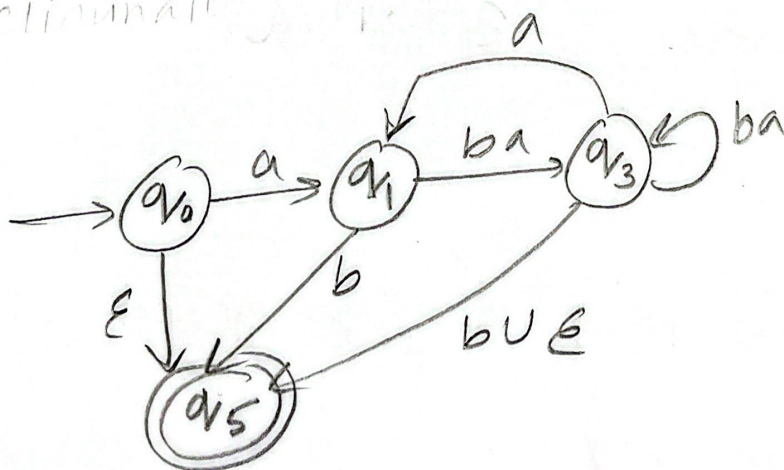
$$q_3 \rightarrow q_4 : bb$$



- eliminating q_4 ,

since q_4 has no outgoing transitions, q_4 will be removed. (As it cannot go to the accepting state.)

- eliminating q_1 :



eliminating q_1 :

$$(q_0, q_3) \times (q_3, q_5)$$

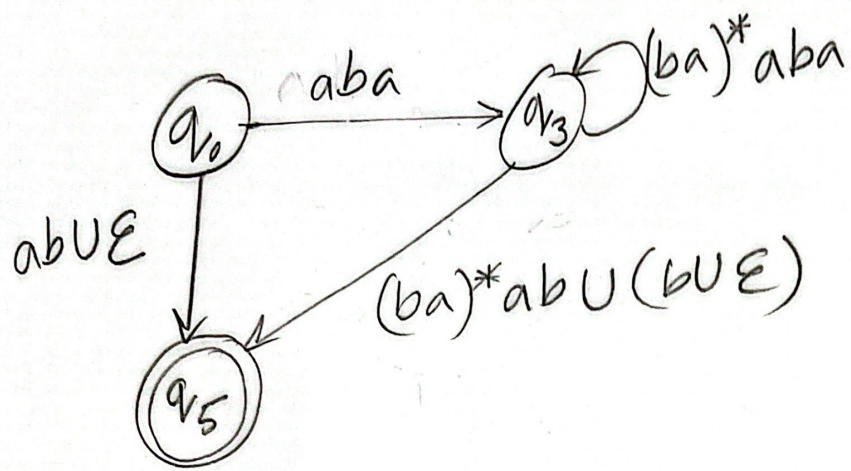
$$(q_0, q_3), (q_0, q_5), (q_3, q_3), (q_3, q_5)$$

$$q_0 \rightarrow q_3: a, ba$$

$$q_0 \rightarrow q_5: ab$$

$$q_3 \rightarrow q_3: (ba)^* a, ba$$

$$q_3 \rightarrow q_5: (ba)^* ab$$



- eliminating q_3

$q_0 \times q_5$

$q_0 \rightarrow q_5 : aba((ba)^*aba)^*(ba)^*abU(bUE)$



$aba((ba)^*aba)^*(ba)^*abU(bUE)U(abUE)$